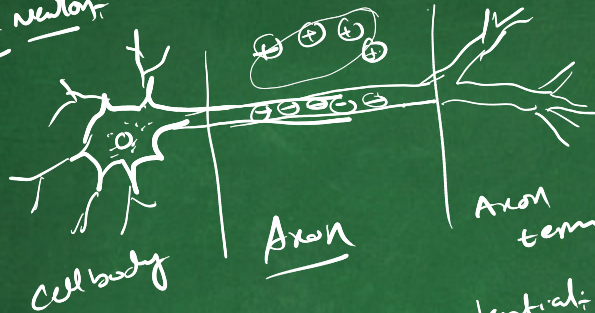


Artificial Neuron:-

Biological neuron:-



Membrane potential:- $\Rightarrow -76\text{mV}$ (at Rest)

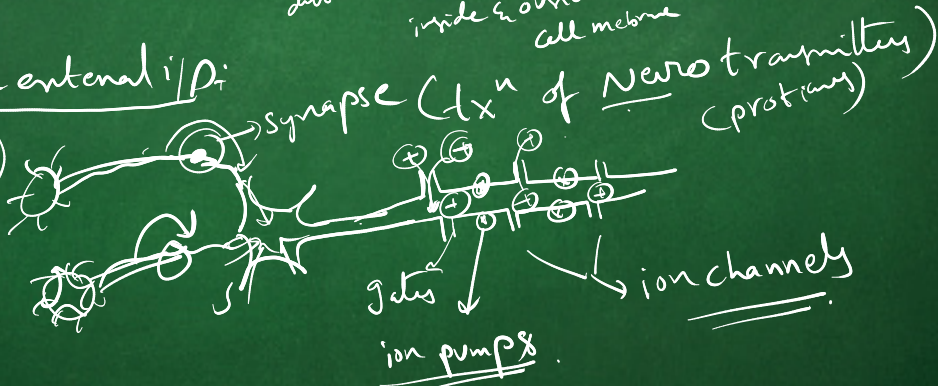
diff. in charge b/w
inside & outside
cell mem.
all mem.

At rest: no i/p

excited with some external i/p:

i/p

(Stimulus)



membrane potential

-40mV
-70mV

IPSP

stimulus

gradient potential

Ion channels $\Rightarrow$ gates $\Rightarrow$ Controlled by neurotransmitter

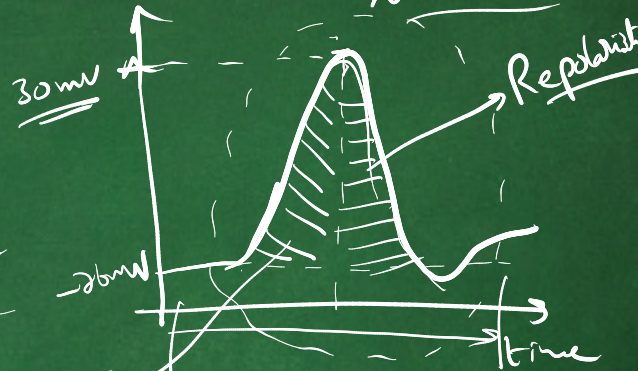
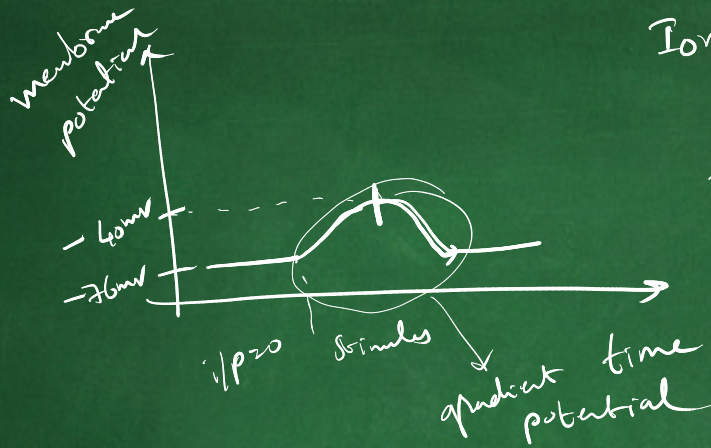
30mV

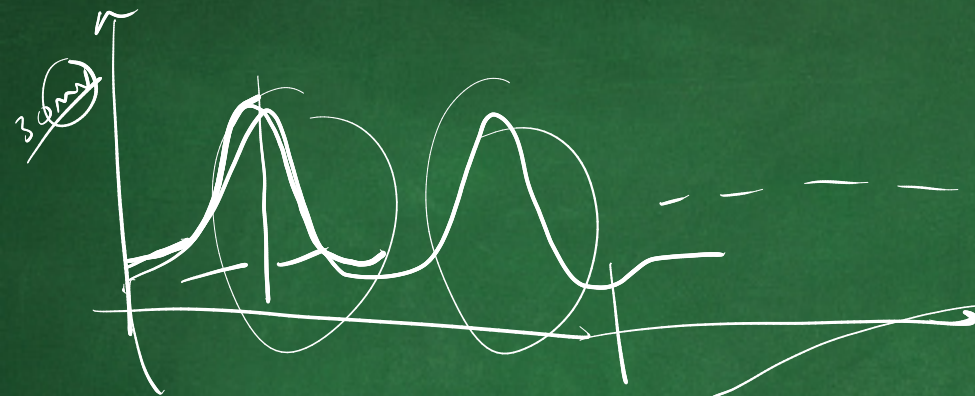
-70mV

Repolarization

Depolarization

Action potential (neuron spike) T_0





0
Spikes/sec

freq $\propto$ stimulus
(strength).

EEG (bands)

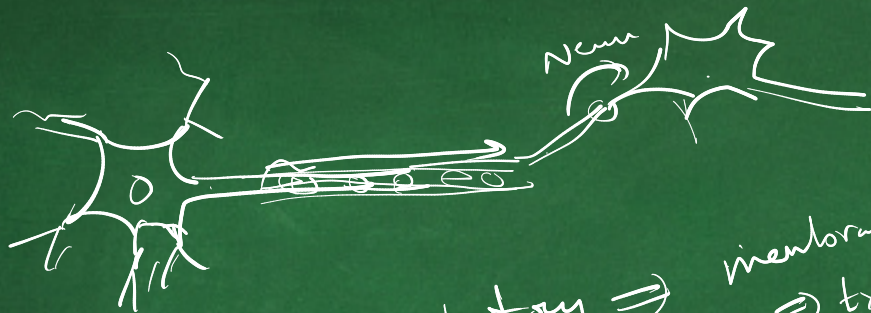
normal

0-4 Hz δ band

4-8 Hz α

8-12 Hz β

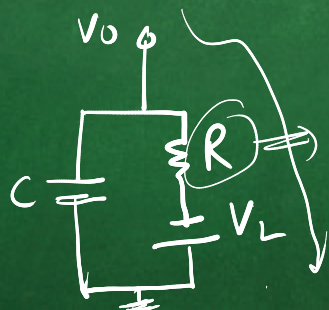
> 12 Hz γ
(high)



Action potential $\Rightarrow$ Excitatory $\Rightarrow$ membrane potential $\uparrow \uparrow$
 $\Rightarrow$ tx or neuro transmits to other neuron.

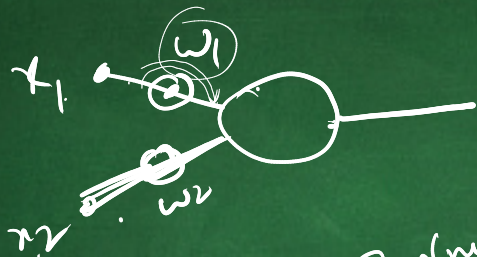
gradient potential $\Rightarrow$ Inhibitory

$\Rightarrow$ membrane potential is low
 $\rightarrow$ no exchange of neuro transmitter.



Ion channels

$$V_{\text{channel}} = (V_0 - V_L)$$



$$net = \underline{w_1 x_1} + w_2 x_2 \quad \checkmark \quad \text{A.P.}$$

$$\text{sgn}(net) = 1 \text{ if } net > 0 \Rightarrow \text{excitatory} \quad 0 = 1$$

$$= 0 \text{ if } net \leq 0 \Rightarrow \text{Inhibitory} \quad \text{AP} \quad 0 = 0$$

↳ McCulloch & Pitts model



$$0 = 1 \text{ or } 0$$

$$0 = \text{sgn}(net)$$

$$net = \sum_{k=1}^K w_k x_k ; 0 = \text{sgn}(net)$$

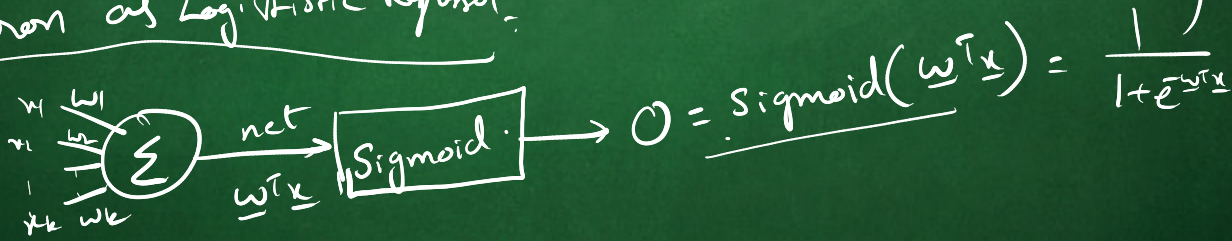
if $\underline{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$; if $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \Rightarrow \underline{w}^T \underline{x} = \underline{x}^T \underline{w} = \sum_{k=1}^n w_k x_k$

vector form:

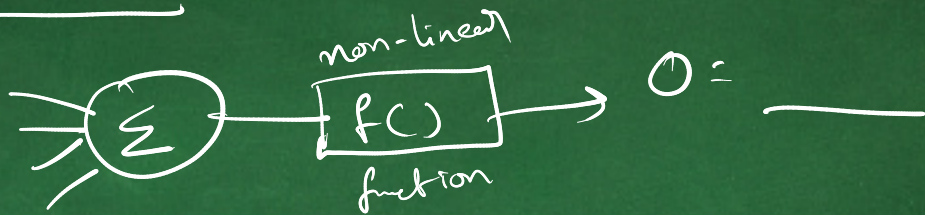
$\text{net} = \underline{w}^T \underline{x}$; $\underline{O} = \text{sgn}(\text{net}) = \text{sgn}(\underline{w}^T \underline{x})$

Logistic Reg: $\hat{y} = \text{Sigmoid}(\underline{w}^T \underline{x}) = \frac{1}{1 + e^{-\underline{w}^T \underline{x}}}$

neuron as Logistic Repr:



Artificial Neuron :



nonlinear function (or) Activation function

- (i) o/p is binary 0 or $1 \Rightarrow f(x) = \text{sign}()$
- (ii) $o/p \in \{0, 1\} \Rightarrow f(x) = \text{sigmoid}()$
- (iii) $o/p \in \{-1, 1\} \Rightarrow f(x) = \text{tanh}()$

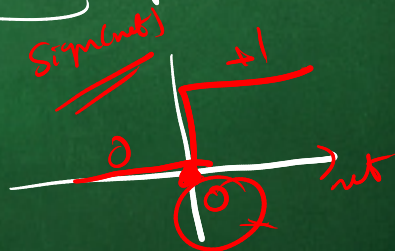
Ex: $\underline{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ $\underline{w} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$. What is the o/p or neural
If $f() = \text{Sign}()$?

$$\text{net} = \underline{w}^T \underline{x} = 1 \times 0 + (-1)(-1) + 2 \times 1 = 3$$

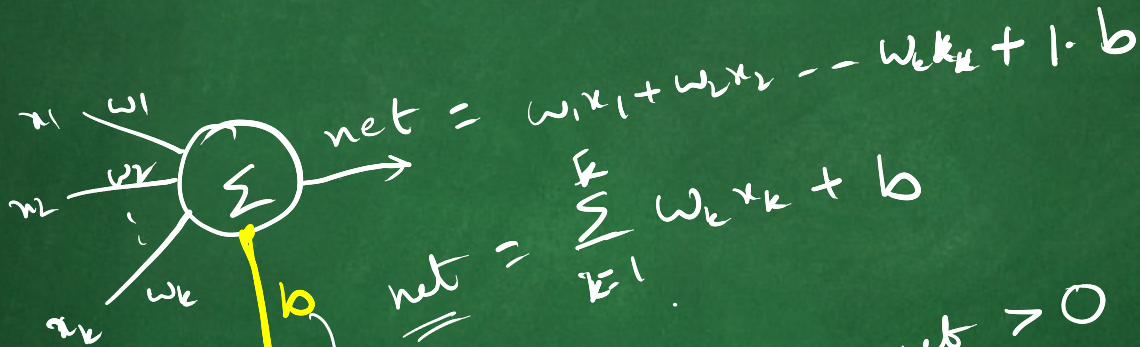
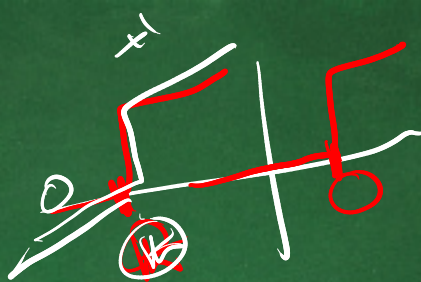
$$o = \text{Sign}(\text{net}) = \text{Sign}(3) = \underline{\underline{+1}}$$

$\text{net} = \underline{w}^T \underline{x} \Rightarrow$ eff no. or neto transmitter.

$$\phi = 1 \quad \text{net} > 0$$
$$= 0 \quad \text{net} \leq 0$$



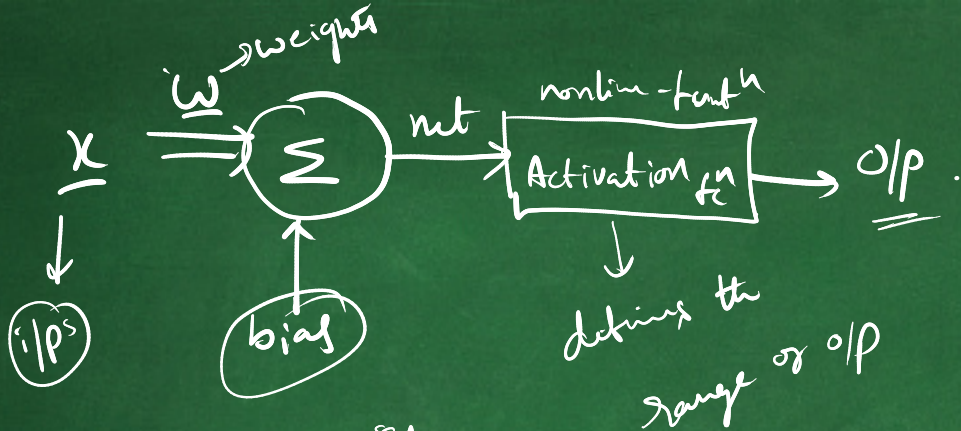
for non-zero reference:
 $O = 1$ if $net > th$
 $= 0$ if $net \leq th$



bias

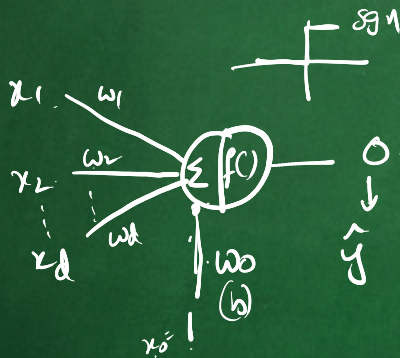
$O = \text{sgn}(net) = 1$

if $net > 0$
 $\sum_{k=1}^K w_k x_k + b > 0$
 $\sum_{k=1}^K w_k x_k > -b$



$$\text{net} = \underline{w}^T \underline{x}$$

$$\frac{\partial \text{net}}{\partial \underline{w}} = \underline{x}$$



$$O = f\left(\sum_{p=0}^d w_p x_p\right) \rightarrow \text{net}; x_0 = 1$$

$$= f(\underline{w}^T \underline{x})$$

$$\underline{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}$$

$$\underline{x} = \begin{bmatrix} x_0 = 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}$$

$$\{x_i, y_i\}_{i=1}^N$$

$$\hat{y}_i = f(\underline{\omega}^T \underline{x}_i) = f(\text{net})$$

$$E = L(y_i, \hat{y}_i) \begin{cases} \text{sq error } \sum_i (y_i - \hat{y}_i)^2 \\ \text{cross entropy } - \left(\sum_i y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right) \end{cases}$$

min E w.r.t $\underline{\omega} \Rightarrow$ Gradient Descent

$$\underline{\omega}^{\text{new}} = \underline{\omega}^{\text{old}} - \eta \frac{\partial E}{\partial \underline{\omega}} \Leftarrow \frac{\partial E}{\partial \underline{\omega}} \text{ (gradient of loss function)}$$

$$\frac{\partial}{\partial \underline{\omega}} L(y, \hat{y}) = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial \text{net}} \times \frac{\partial \text{net}}{\partial \underline{\omega}}$$

$$L(y, \hat{y}) \leftarrow \hat{y}$$

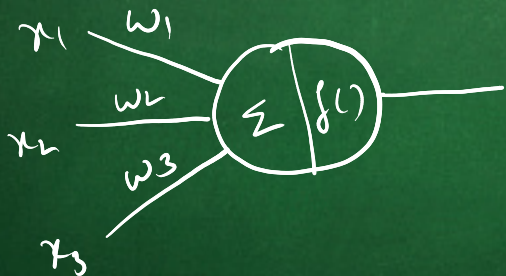
$$\hat{y} \leftarrow \text{net}$$

$$\text{net} \leftarrow \underline{\omega}$$

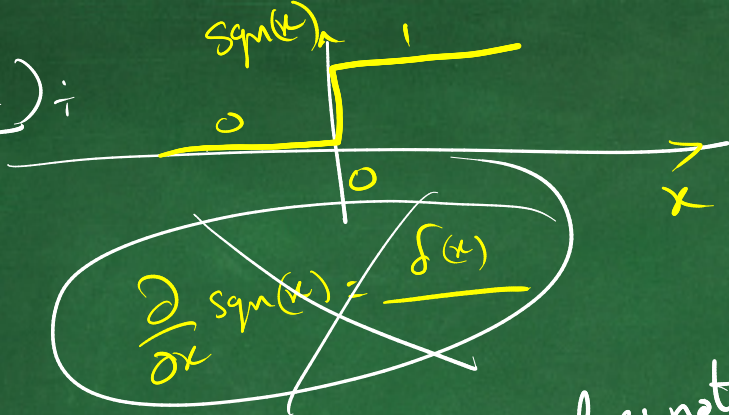
$$\frac{\partial L}{\partial \underline{\omega}} = L'(y, \hat{y}) \cdot \underline{f'(\text{net}) \cdot x}$$

$$\text{if } L = (y - \hat{y})^2$$

$$\frac{\partial L}{\partial \underline{\omega}} = 2(y - \hat{y}) \cdot \underbrace{f'(\text{net})}_{\omega^T x} \cdot \underline{x}$$



Sigm(x) :



if $f(x) = \text{sgn}(x) \Rightarrow f'(x)$ does not exist.

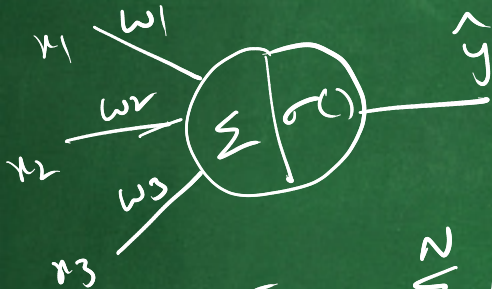
Sigmoid : $f(x) \Rightarrow \frac{1}{1 + e^{-x}}$

$f'(x) = \frac{1}{(1 + e^x)^2} \cdot e^{-x}$

$$\sigma(x)(1 - \sigma(x)) = \frac{1}{1 + e^x} \left(1 - \frac{1}{1 + e^x} \right)$$

$$= \frac{e^{-x}}{(1 + e^x)^2}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



find update equation for w_2

$\{x^i, y^i\}_{i=1}^N$

$$E = \sum_{i=1}^N (y^i - \hat{y}^i)^2$$

$$\underline{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \end{bmatrix}$$

$$w_2^{\text{new}} = w_2^{\text{old}} - \eta \frac{\partial E}{\partial w_2}$$

$$\frac{\partial E}{\partial \underline{w}} = \begin{bmatrix} \frac{\partial E}{\partial w_0} \\ \frac{\partial E}{\partial w_1} \\ \vdots \end{bmatrix}$$

$$\frac{\partial E}{\partial \omega_2} = \frac{\partial E}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial \omega_2}$$

$$= \frac{\partial (y - \hat{y})^2}{\partial \hat{y}} \times \frac{\partial \sigma(z)}{\partial z} \times \frac{\partial \sum_{p=0}^d \omega_p x_p}{\partial \omega_2}$$

$$\text{net} = z = \sum_{p=1}^d \omega_p x_p$$

in the code

$$z^{(i)} = \omega^T x^{(i)}$$

$$\omega_0 x_0 + \omega_1 x_1 + \omega_2 x_2 + \dots$$

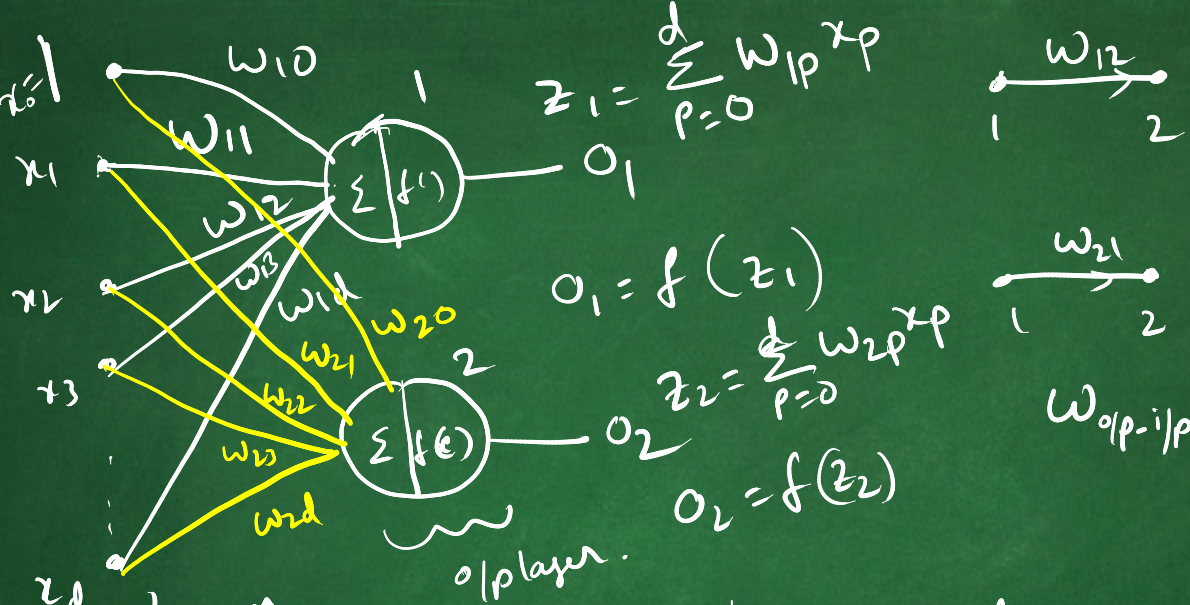
$$\frac{\partial E}{\partial \omega_2} = \sum_{i=1}^N 2 (y^{(i)} - \hat{y}^{(i)}) \times \sigma(z^{(i)}) (1 - \sigma(z^{(i)})) \times x_2^{(i)}$$

update $\omega_0, \omega_1, \omega_2, \dots$

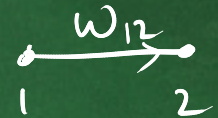
$$\frac{\partial E}{\partial w_k} = 2 \sum_{i=1}^N (y^{(i)} - \hat{y}^{(i)}) \cdot \sigma(z^{(i)} (1 - \sigma(z^{(i)}))) \times x_k^{(i)}$$

$$E = y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})$$

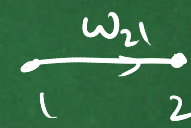
$$\frac{\partial E}{\partial \hat{y}^{(i)}} = \left(\frac{y^{(i)}}{\hat{y}^{(i)}} - \frac{1 - y^{(i)}}{1 - \hat{y}^{(i)}} \right)$$



$$z_1 = \sum_{p=0}^d w_{1p} x_p$$



$$o_1 = f(z_1)$$



$$z_2 = \sum_{p=0}^d w_{2p} x_p$$

$$w_{o/p-i/p}$$

$$o_2 = f(z_2)$$

o/p layer.

Neural Network

$$o_1 = f(z_1) = f\left(\sum_{p=0}^d w_{1p} x_p\right)$$

$$z_1 = \underline{w}_1^T \underline{x}$$

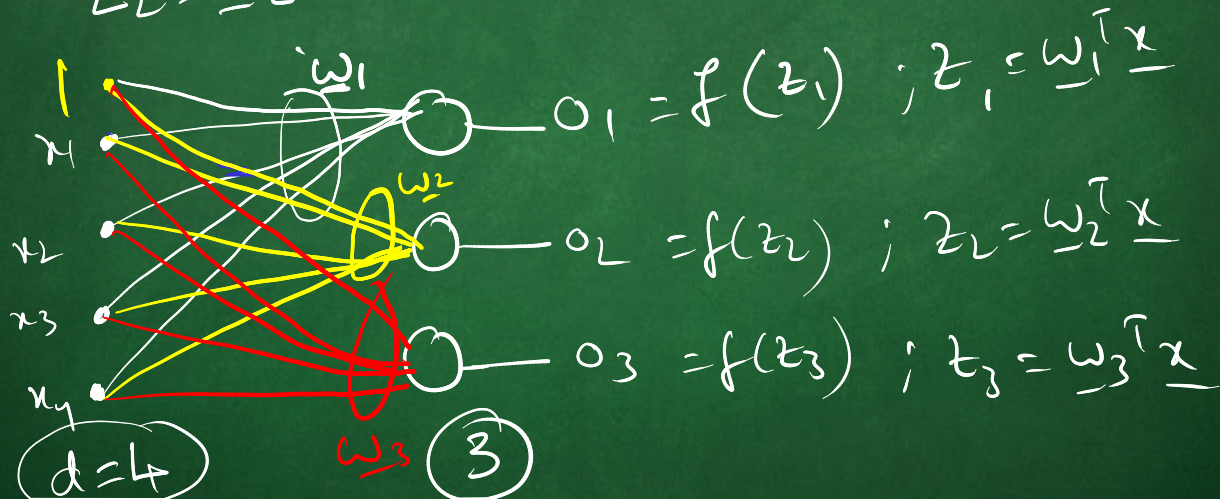
"Single" layer

$$\text{when } \underline{w}_1 = \begin{bmatrix} w_{10} \\ w_{11} \\ w_{12} \\ \vdots \end{bmatrix}$$

$$\underline{x} = \begin{bmatrix} x_0 = 1 \\ x_1 \\ x_2 \\ \vdots \end{bmatrix}$$

$$\underline{w}_2 = \begin{bmatrix} w_{20} \\ w_{21} \\ w_{22} \\ \vdots \end{bmatrix}$$

$$z_2 = \underline{w}_2^T \underline{x}$$



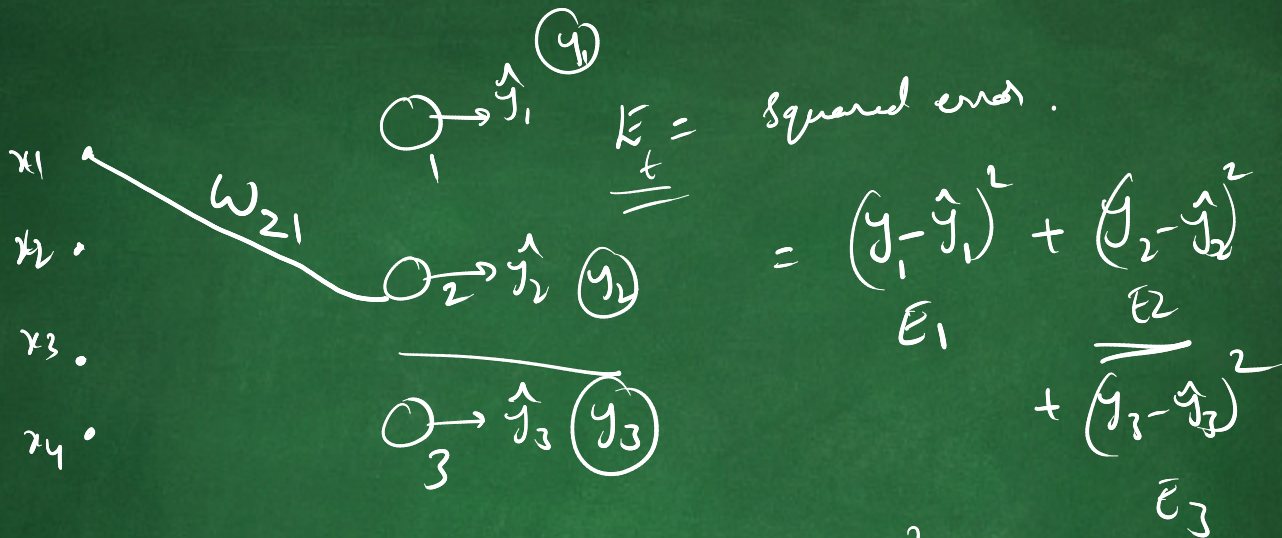
$$\underline{f(\cdot) = \text{Sigmoid}} \Rightarrow \hat{y}_1, \hat{y}_2, \hat{y}_3 \in [0, 1]$$

↳ softmax

○ $f(\cdot) = \text{softmax}$ for multi-class classification

○ $\text{softmax}(\underline{w_k^T x}) = \frac{e^{\underline{w_k^T x}}}{\sum_{k=1}^K e^{\underline{w_k^T x}}}$

○



$$E = E_1 + E_2 + E_3 = \sum_{k=1}^3 (y_k - \hat{y}_k)^2$$

what is the update equation for " w_{21} "

$$w_{21}^{nw} = w_{21}^{old} - \eta \frac{\partial E}{\partial w_{21}}$$

$$\frac{\partial E}{\partial w_{21}} = \frac{\partial [E_1 + E_2 + E_3]}{\partial w_{21}} = 0 + \frac{\partial E_2}{\partial w_{21}} + 0$$

$$= \frac{\partial E_2}{\partial w_{21}} \quad \left| \quad \begin{aligned} E_2 &= (y_2 - \hat{y}_2)^2 \\ \hat{y}_2 &= \sigma(z_2) \\ z_2 &= \sum_{p=0}^d w_{2p} x_p \end{aligned} \right.$$

$$\frac{\partial E_2}{\partial w_{21}} = \frac{\partial E_2}{\partial \hat{y}_2} \times \frac{\partial \hat{y}_2}{\partial z_2} \times \frac{\partial z_2}{\partial w_{21}}$$

$$w_{20} + \underbrace{w_{21} x_1}_{+} + w_{22} x_2 + \dots$$

$$\frac{\partial E}{\partial w_{21}} = 2(y_2 - \hat{y}_2) \times f'(z_2) \times x_1$$

$$\frac{\partial E}{\partial w_{21}} = 2(y_2 - \hat{y}_2) \times f'(z_2) \times x_1$$

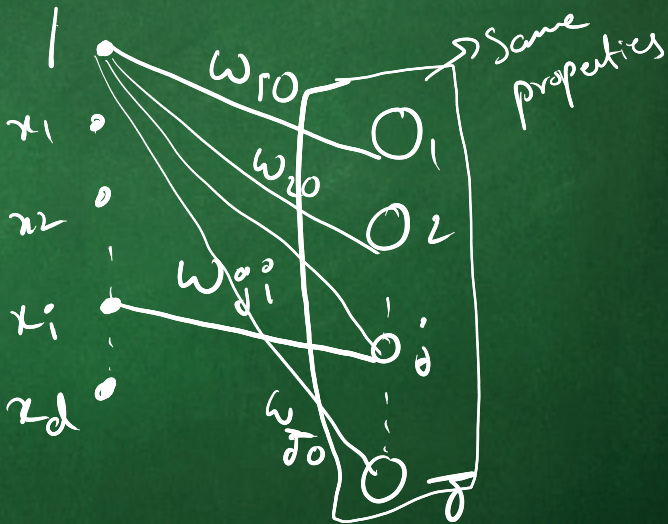
Generalized gradient

$$\frac{\partial E}{\partial w_{ij}} = \sum_{n=1}^N (y_i^{(n)} - \hat{y}_i^{(n)}) \times f'(z_i^{(n)}) \times x_j^{(n)}$$

$$w_{ij}^{new} \leftarrow w_{ij}^{old} - \eta \frac{\partial E}{\partial w_{ij}}$$

no. of parameters in a ANN w/o (ANN)

i/p $\Rightarrow$ d-dimensional
 no. of neurons $\Rightarrow$ J
 in o/p layer



one neuron : d inputs $\Rightarrow d$ weights.

+ 1 bias

$\Rightarrow$ no. of params = $d+1$

total no. of params in single layer n/w with
 d -dim i/p in J neurons $\Rightarrow J(d+1)$

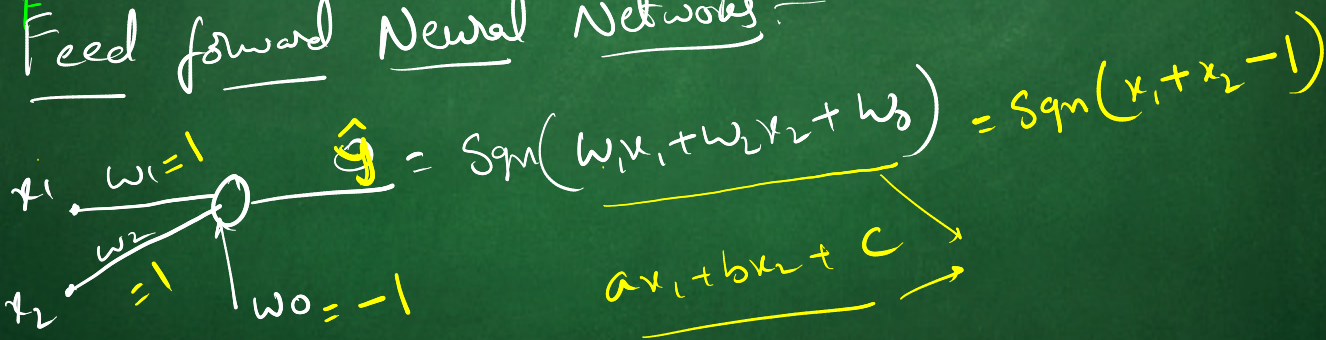
need to find gradients for each of $J(d+1)$

no. of params $\Rightarrow w_{ji} \leftarrow w_{ji} - \eta \frac{\partial E}{\partial w_{ji}}$

$$\underline{\underline{w_{ji}}} \Rightarrow \forall j=1, 2, \dots, J \rightarrow J \quad \left. \begin{matrix} i=0, 1, 2, \dots, d \rightarrow d+1 \end{matrix} \right\} J(d+1)$$

$$\frac{\partial E}{\partial w_{ji}} = \sum_{n=1}^N \frac{\partial E}{\partial \hat{y}_j^{(n)}} \cdot f'(\hat{z}_j^{(n)}) \cdot x_i^{(n)}$$

Feed forward Neural Networks:



$$\hat{g} = \text{Sgm}(w_1 x_1 + w_2 x_2 + w_0) = \text{Sgm}(x_1 + x_2 - 1)$$

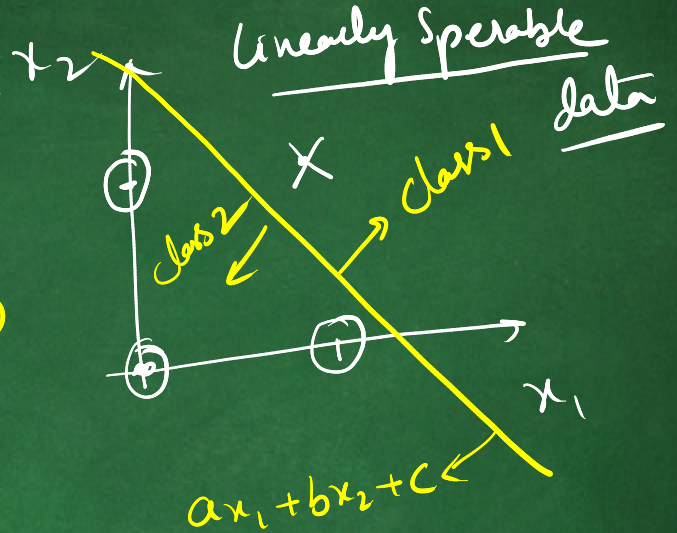
$$\underline{ax_1 + bx_2 + c}$$

ex:

AND

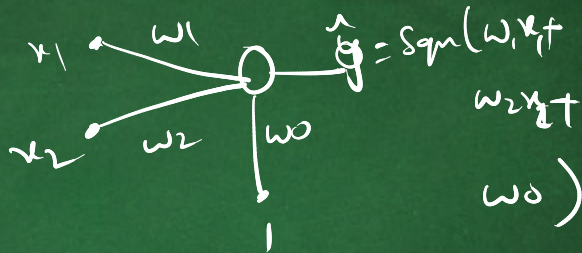
x_1	x_2	$\hat{y}$
0	0	0
1	0	0
0	1	0
1	1	1

$$\hat{y} = \text{sqm}(x_1 + x_2 - 1)$$

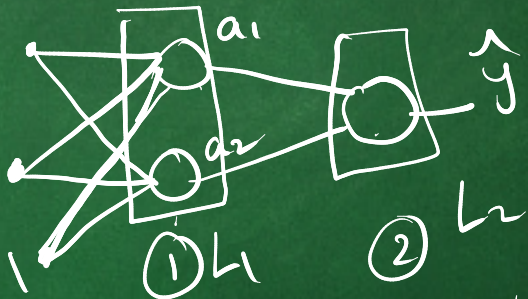
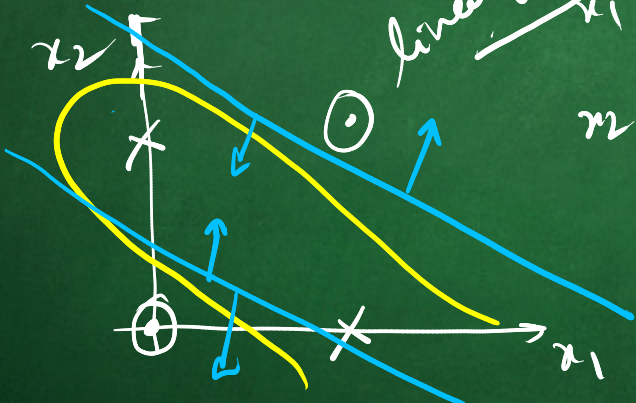


not

x_1	x_2	$\odot$	$\hat{y}$
0	0	0	
1	0	1	
0	1	1	
1	1	0	

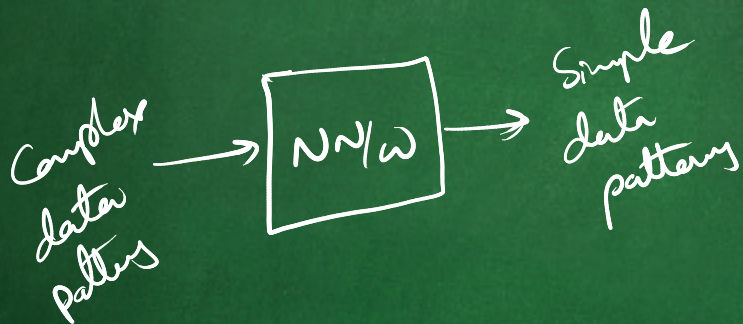


linearly NOT separable

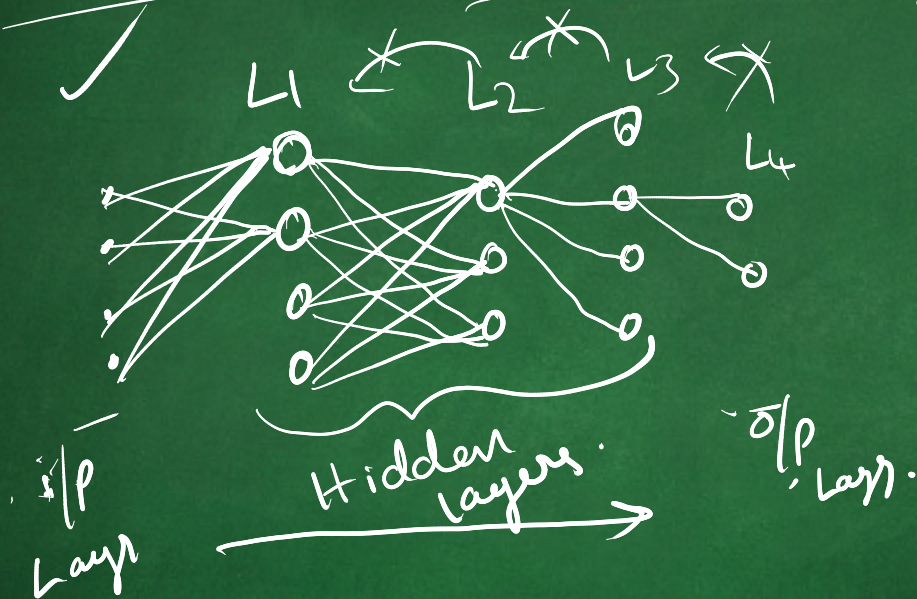


2-Layer Neural N/w

$\{x_1, x_2\} \Rightarrow \text{linearly Not separable}$
 $\Downarrow \text{Transform}$
 $\{a_1, a_2\} \Rightarrow \text{linearly Separable.}$



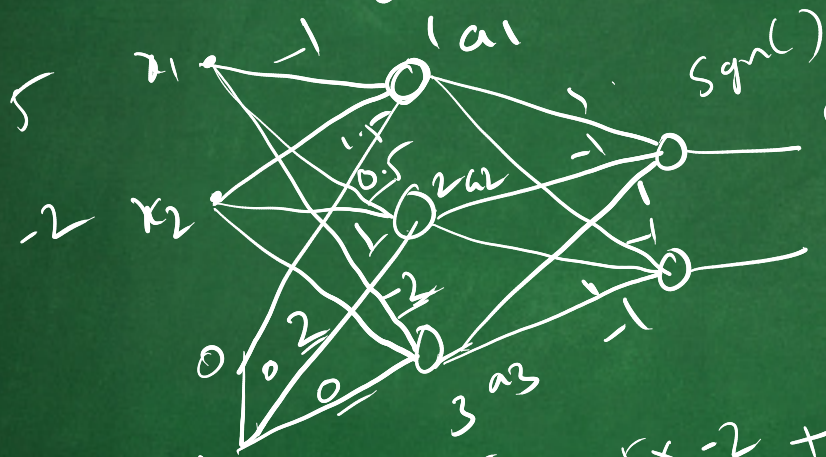
Multi Layer Feed Forward Neural n/w:



Ex

~~with~~
 $\text{sgn}()$

$O_1 =$



$$O_1 = \text{sgn}(1 \cdot 0 - 1 \cdot 1 + 1 \cdot 0) \\ = \text{sgn}(-1) = 0$$

$$O_2 = \text{sgn}(-1 \cdot 0 + 1 \cdot 1 - 1 \cdot 0) \\ = \text{sgn}(1) = 1$$

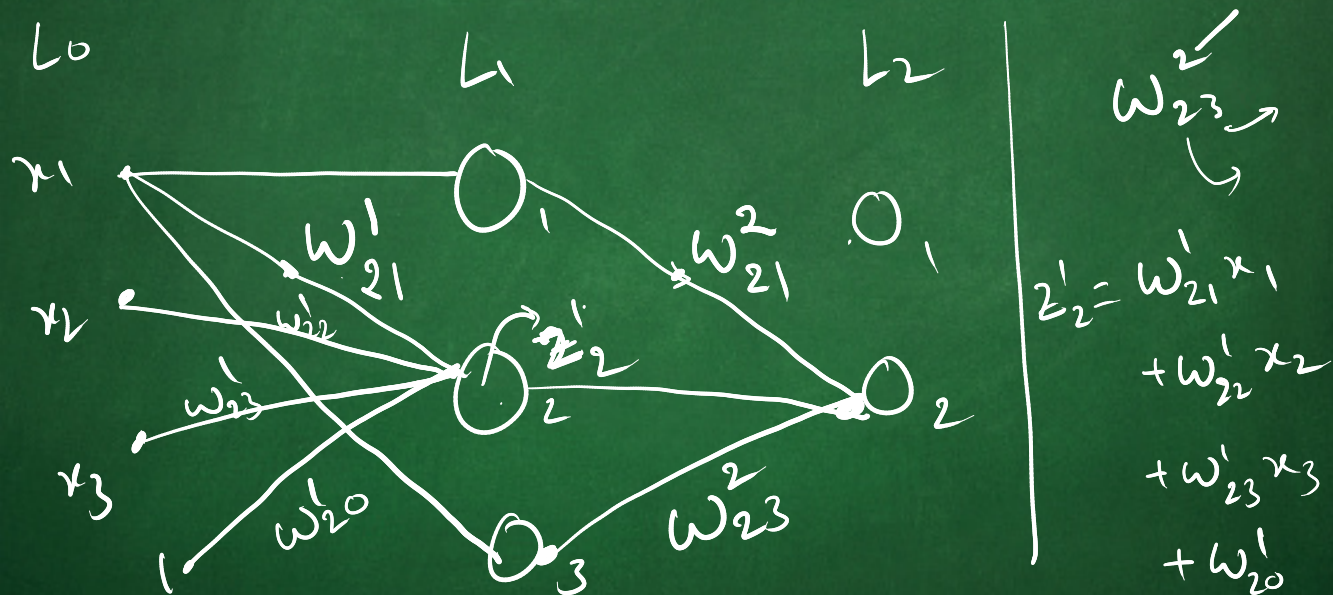
$$= \text{sgn}(-5 - 3 + 0)$$

$$a_1 = \text{sgn}(-1 \times 5 + 1.5 \times -2 + 0)$$

$$a_2 = \text{sgn}(0.5 \times 5 + 1 \times -2 + 0) \\ = 0 \\ = \text{sgn}(0.5) = 1$$

$$a_2 = \underline{1}; a_3 = \underline{0}; o_1 = \underline{0}; o_2 = \underline{1}$$

$$a_1 = \underline{0} \quad a_3 z \sin(5 \times 2 + (-2) 2 + 0) = \sin(-14) = 0$$



Notation : FFNN

$l \rightarrow$ layer index

$W_{ji} \rightarrow$ i/p node index

$\rightarrow$ o/p node index

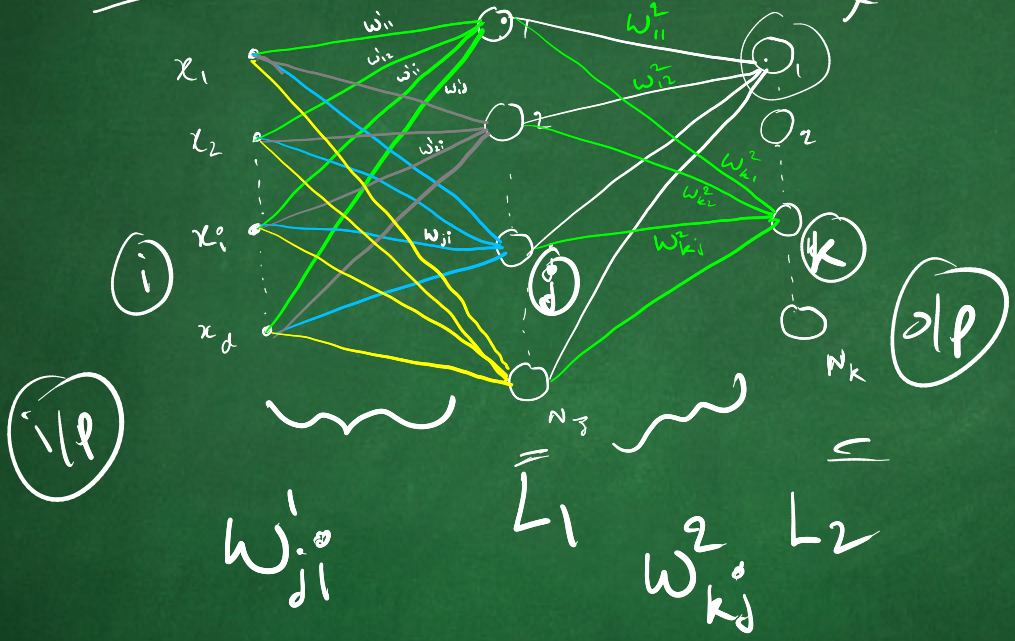
net

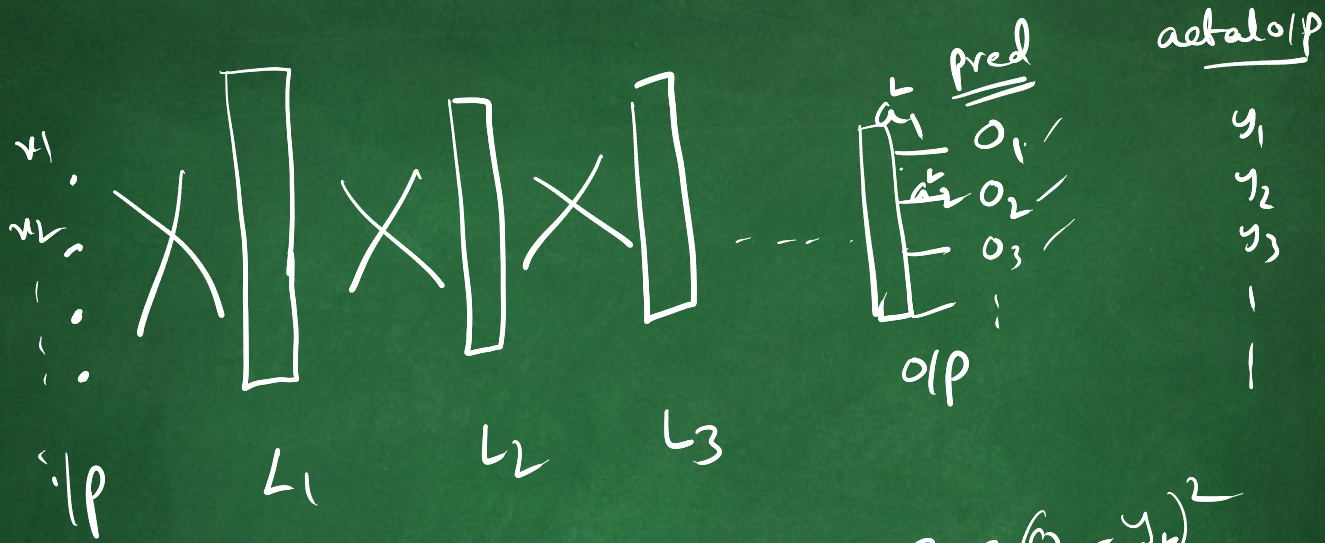
$z_j^l \rightarrow$ layer index
 $\rightarrow$ node index

$\Rightarrow$ net value of j^{th} node
in layer ' l '

o/p of j^{th} neuron in layer ' l ' : $a_j^l = f(z_j^l) = \sigma(z_j^l)$

2-layer FF NN: Hidden layer.





Error:

$$e_1 = (o_1 - y_1)^2$$

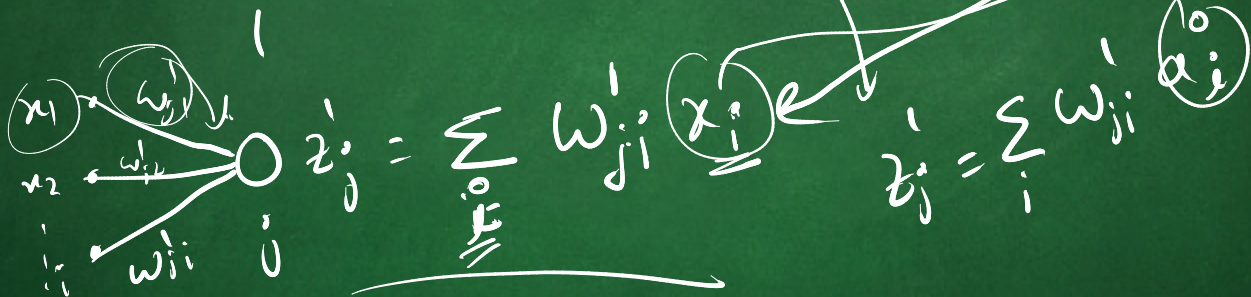
$$e_k = (o_k - y_k)^2$$

$$e_2 = (o_2 - y_2)^2$$

$$E = e_1 + e_2 + \dots$$

o/p of a node j in layer l

$$a_j^l = \sigma(z_j^l) ; z_j^l = \sum_i w_{ji}^l a_i^{l-1}$$



$$\{ \underbrace{\omega_{ij}^l}_{\text{given}}, a_j^l, z_j^l \} \quad \forall l=1, 2, \dots, \underline{L}$$

o/p of L^{th} is : $a_k^L ; k=1, 2, \dots, \underline{N_k}$

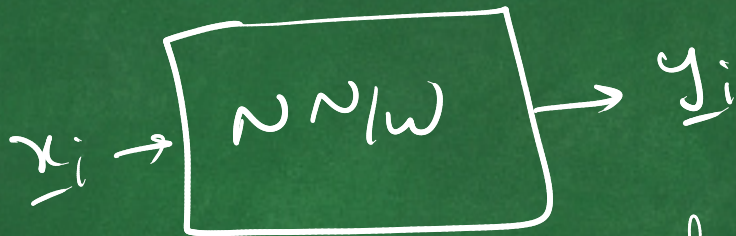
$\downarrow$
final (a) o/p layer

$$O_k = \underline{a_k^L} = \sigma(z_k^L)$$

$$z_k^L = \sum_i \omega_{ki}^L \cdot a_i^{L-1}$$

$$\{\underline{x}_i, \underline{y}_i\}_{i=1}^N \Rightarrow \underline{y}_i = f(\underline{x}_i)$$

$\mathcal{F}$



prob
minimize $E(o_i, y_i)$ w.r.t weight $\boxed{w_{ji}^2}$

one error $\overset{\text{pred}}{o}, \overset{\text{act}}{y}$

$$E = \underbrace{(o - y)^2}_{\substack{\downarrow \\ \text{Squared error}}} \quad (o) - \underbrace{y \log o}_{\substack{\downarrow \\ \text{entropy}}} \quad \overset{\text{pred}}{\text{pred}(y)}$$

$S \log \frac{1}{p(s)}$

$-\log p(s)$

entropy

$$J = J_1 + J_2 + \dots + J_N$$

$$= (o_1 - y_1)^2 + (o_2 - y_2)^2 + \dots + (o_N - y_N)^2$$

$$\frac{\partial \mathcal{J}}{\partial \omega} = \frac{\partial \mathcal{J}_1}{\partial \omega} + \frac{\partial \mathcal{J}_2}{\partial \omega} + \dots + \frac{\partial \mathcal{J}_N}{\partial \omega}$$

Solⁿ, Gradient descent.

$$\left(\omega_{ji}^l \right)^{\text{new}} = \left(\omega_{ji}^l \right)^{\text{old}} - \eta \boxed{\frac{\partial \mathcal{J}}{\partial \omega_{ji}^l}}$$

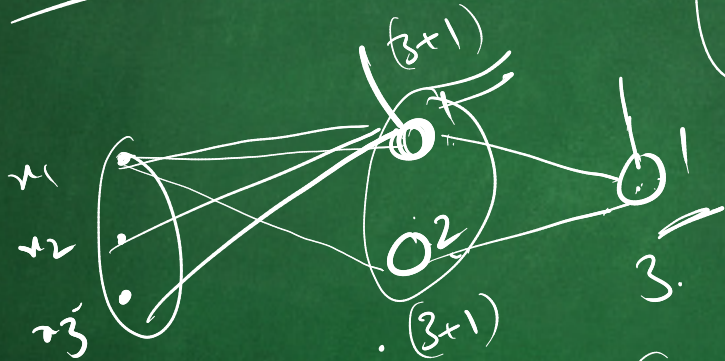
for every i, j, l

↓
Gradient of loss

Finding the Gradients $\left(\frac{\partial \mathcal{L}}{\partial w_{ji}^l} \right)$ for all weights in all layers

no. of parameters

# L1	# L2	# L3



$$1 \cdot (2+1) = \underline{\underline{3}}$$

$$1 \cdot (4) \times (2) = \underline{\underline{8}} + \underline{\underline{3}} = \underline{\underline{11}} \text{ params.}$$

$$\# L_\ell = N_\ell (N_{\ell-1} + 1) \rightarrow \text{bias}$$

no. of nodes in ℓ^{th} layer.

nodes in $(\ell-1)^{\text{th}}$ layer

$$\text{total no. of param} = \sum_{\ell=1}^L \# L_\ell$$

3dim
fil



4



$$4 \times (3+1) = 16$$



6



$$6(4+1) = 30$$



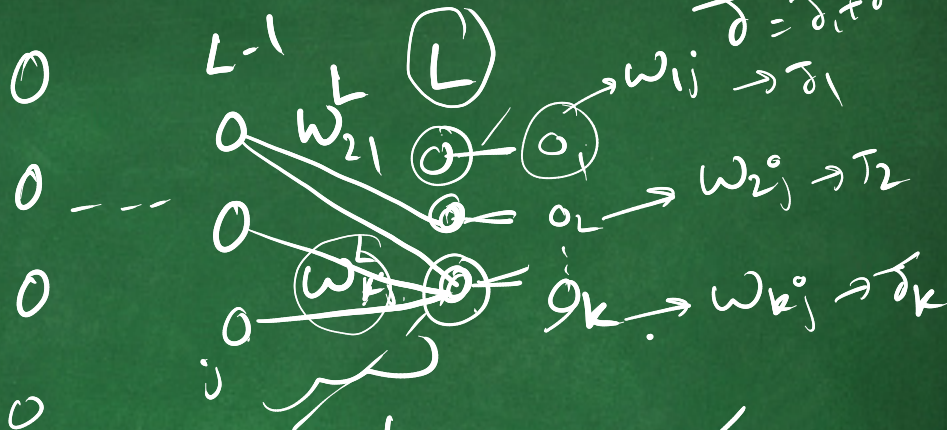
3



$$3(6+1) = \underline{21}$$



67



$$\frac{\partial \mathcal{J}}{\partial w_{kj}^L} = \frac{\partial \mathcal{J}_1}{\partial w_{kj}^L} + \frac{\partial \mathcal{J}_2}{\partial w_{kj}^L} + \dots + \frac{\partial \mathcal{J}_K}{\partial w_{kj}^L} + \dots$$

$$\frac{\partial \mathcal{J}}{\partial \omega_{kj}^L} = \frac{\partial \mathcal{J}_k}{\partial \omega_{kj}^L}$$

$$= \frac{\partial \mathcal{J}_k}{\partial a_k^L} \times \frac{\partial a_k^L}{\partial z_k^L} \times \frac{\partial z_k^L}{\partial \omega_{kj}^L}$$

$$= 2(a_k - y_k) \times \sigma'(z_k^L) \times a_j^{L-1}$$

$$\mathcal{J}_k = (O_k - y_k)^2$$

$$O_k = a_k^L$$

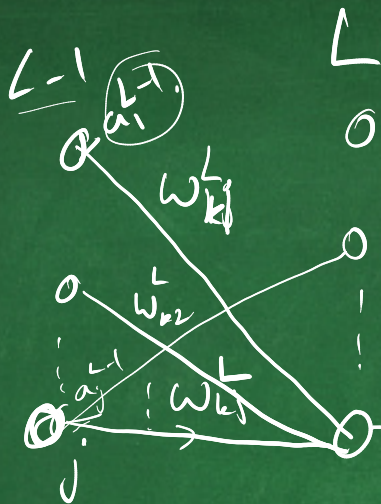
$$a_k^L = \sigma(z_k^L)$$

$$z_k^L = \sum_j \omega_{kj}^L a_j^{L-1}$$

$$\frac{\partial \mathcal{J}}{\partial w_{kj}^L} = \frac{\partial \mathcal{J}_k}{\partial w_{kj}^L} = \frac{\partial \mathcal{J}_k}{\partial a_k^L} \frac{\partial a_k^L}{\partial z_k^L} \times a_j^{L-1}$$

$$\frac{\partial \mathcal{J}}{\partial w_{kj}^L} = \delta_k^L \cdot a_j^{L-1}$$

① ~~$\delta_k^L = \frac{\partial \mathcal{J}}{\partial a_k^L} \frac{\partial a_k^L}{\partial z_k^L} = \frac{\partial \mathcal{J}}{\partial a_k^L} \sigma'(z_k^L)$~~



$$\frac{\partial \mathcal{J}}{\partial w_{kj}^L} = \delta_k^L a_j^{L-1}$$

$$o_k = a_k^L = \sigma(z_k^L)$$

$$z_k^L = \sum_j w_{kj}^L a_j^{L-1}$$

$$\frac{\partial \mathcal{J}}{\partial w_{11}^L} = \delta_1^L \cdot a_1^{L-1}$$

$$\delta_1^L = \frac{\partial \mathcal{J}}{\partial a_1^L} \sigma'(z_1^L)$$

$$\delta_1^L = 2(o_1 - y_1) (o_1)(1-o_1)$$

$$= 2(o_1 - y_1) \times \frac{a_1^L(1-a_1^L)}{o_1(1-o_1)}$$

$$\delta_2^L = 2(o_2 - y_2) o_2(1-o_2)$$

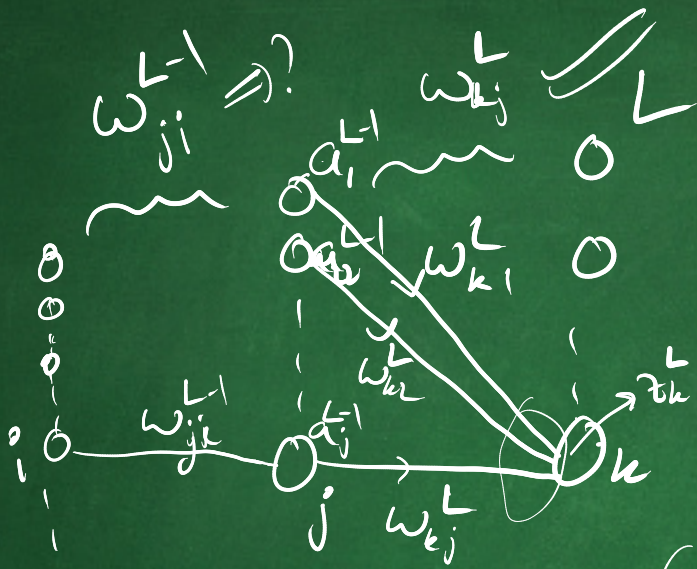
error term

o/p of prev layer.

$$\frac{\partial \bar{J}}{\partial w_{21}^L} = \delta_2^L a_1^{L-1}$$

grad = error $\times$ o/p of prev layer

(w.r.t weights)



$$\frac{\partial \mathcal{L}}{\partial w_{kj}^L} = \delta_k^L a_j^{L-1}$$

$$\frac{\partial \mathcal{L}}{\partial w_{ji}^{L-1}} = ?$$

$$(w_{ji}^{L-1})^{\text{new}} = (w_{ji}^{L-1})^{\text{old}} - \eta \frac{\partial \mathcal{L}}{\partial w_{ji}^{L-1}}$$

$$z_k^L = \sum_j w_{kj}^L a_j^{L-1}$$

$$\delta = \delta_1 + \delta_2 + \dots + \delta_k = \sum_k \delta_k$$

$$\frac{\partial \delta}{\partial \omega_{ji}^{L-1}} = \sum_k \frac{\partial \delta_k}{\partial \omega_{ji}^{L-1}} \Rightarrow (1)$$

$$\frac{\partial \delta_k}{\partial \omega_{ji}^{L-1}} = \frac{\partial \delta_k}{\partial a_k^L} \times \frac{\partial a_k^L}{\partial z_k^L} \times \frac{\partial z_k^L}{\partial \omega_{ji}^{L-1}} \rightarrow (2)$$

$\delta_k = (a_k^L - y_k)^2$
 $a_k^L = \sigma(z_k^L)$
 $\frac{\partial \delta_k}{\partial a_k^L} = 2(a_k^L - y_k)$
 $\frac{\partial a_k^L}{\partial z_k^L} = \sigma'(z_k^L)$
 $\frac{\partial z_k^L}{\partial \omega_{ji}^{L-1}} = ?$

$$z_k^L = \sum_j w_{kj}^L \underline{a_j^{L-1}}$$

$$\frac{\partial z_k^L}{\partial w_{ji}^{L-1}} =$$

$$z_k^L = \sum_q w_{kq}^L \sigma \left(\sum_p w_{qp}^{L-1} a_p^{L-2} \right) \quad a_j^{L-1} = \sigma(z_j^{L-1})$$
~~$$\underline{z_k^L} = \sum_i \underline{w_{ji}^{L-1}} \underline{a_i^{L-2}}$$~~

$$\frac{\partial}{\partial \omega_{ji}^{L-1}} \left[\sum_q \omega_{kq}^L \sigma \left(\sum_p \omega_{qp}^{L-1} a_p^{L-2} \right) \right]$$

$$\Downarrow$$

$$q \neq j \quad \frac{\partial}{\partial \omega_{ji}^{L-1}} () = 0$$

$$q=j \quad \frac{\partial}{\partial \omega_{ji}^{L-1}} \left[\omega_{kj}^L \sigma \left(\sum_p \omega_{jp}^{L-1} a_p^{L-2} \right) \right]$$

$$\Rightarrow w_{kj}^L \sigma'(z_j^{L-1}) \frac{\partial}{\partial w_{ji}^{L-1}} \left(\sum_p w_{jp}^{L-1} a_p^{L-2} \right)$$

$$\downarrow p=1$$

$$\frac{\partial z_k^L}{\partial w_{ji}^{L-1}} \Rightarrow w_{kj}^L \sigma'(z_j^{L-1}) a_i^{L-2} \rightarrow (3)$$

$$\frac{\partial \delta_k}{\partial w_{ji}^{L-1}} = \frac{\partial \delta_k}{\partial a_k^L} \times \frac{\partial a_k^L}{\partial z_k^L} \frac{\partial z_k^L}{\partial w_{ji}^{L-1}} \rightarrow (2)$$

$\underbrace{\frac{\partial \delta_k}{\partial a_k^L} \times \frac{\partial a_k^L}{\partial z_k^L}}_{\sigma'(z_k^L)} \quad L \quad (?)$

$$\frac{\partial J}{\partial w_{ji}^{L-1}} = \sum_k \frac{\partial J_k}{\partial w_{ik}^{L-1}} \downarrow$$

$$= \sum_k \underbrace{\frac{\partial J_k}{\partial a_k^L} \frac{\partial a_k^L}{\partial z_k^L}}_{\delta_k^L} \left[w_{kj}^L \sigma'(z_j^{L-1}) a_i^{L-2} \right]$$

$$\frac{\partial J}{\partial w_{ji}^{L-1}} = \left[\underbrace{\sum_k w_{kj}^L \delta_k^L}_{\delta_j^{L-1}} \underbrace{\sigma'(z_j^{L-1})}_{\delta_j^{L-1}} \right] a_i^{L-2}$$

$$\frac{\partial \mathcal{J}}{\partial w_{ji}^{L-1}}$$

$$= \delta_j^{L-1} a_i^{L-2}$$

grad = max o/p for min

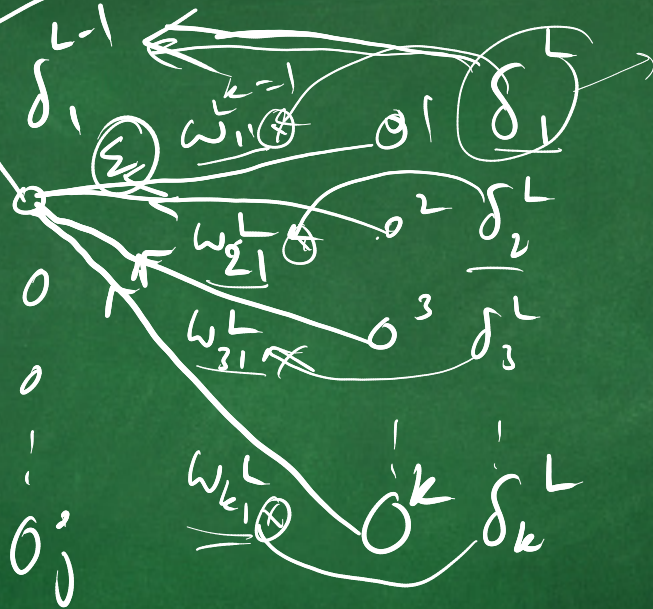
$$= (w_{ij}^L \delta_i^L + w_{2j}^L \delta_2^L + \dots) \sigma'(z_j^{L-1})$$

$$\delta_j^{L-1} = \sum_k w_{kj}^L \delta_k^L \sigma'(z_j^{L-1})$$

$$\delta_k^L = \frac{\partial \mathcal{J}}{\partial a_k^L} \sigma'(z_k^L)$$

$$\frac{1}{\sigma(2_1)} \times \sum_k \omega_{k1}^L \delta_k^L \Rightarrow \delta_1^{L-1}$$

$$\frac{2(\delta_1 - y_1)}{0.4 - 0.1}$$



o/p

o/p

$$\delta_j^{L-1} = \left[\sum_k \omega_{kj}^L \delta_k^L \right] \sigma'(z_j^{L-1})$$

Back propagation of error.

for l^{th} layer = $l=1, 2, 3, \dots, \underline{L-1}$,

$L=l+1$

$\underline{L-1=l}$

$$\underline{\delta_q^l} = \sum_p \omega_{pq}^{l+1} \delta_p^{l+1} \times \sigma'(z_q^l)$$

$$\frac{\partial \mathcal{J}}{\partial w_{qr}^l} = \delta_q^l a_r^{l-1};$$

gradient of
error or Loss($\mathcal{J}$)
w.r.t any
weight in
any layer.

EBP algorithm:

rec eqnth: (1) $\delta_k^L = \frac{\partial \mathcal{J}}{\partial a_k^L} \sigma'(z_k^L)$

$l = L-1$
 $l-2$
 $\vdots$

(2) $\delta_q^l = \sum_p \omega_{qp}^{l+1} \delta_p^{l+1} \sigma'(z_q^l)$

(3) $\frac{\partial \mathcal{J}}{\partial \omega_{qr}^l} = \delta_q^l a_r^{l-1}$

$$④ \quad (w_{qr}^l)^{\text{new}} \leftarrow (w_{qr}^l)^{\text{old}} - \eta \frac{\partial \bar{J}}{\partial w_{qr}^l}$$

EBP Algorithm:

① Initialize $(w_{qr}^l)_{l=1}^L$ with Random Values
for $i=1$ to N
given $x^{(i)}$, $y^{(i)}$

② find $z_q^l, a_q^l, \bar{J}, \delta_q^l$ from ① & ②

$\forall l=1, 2, \dots, L$

{ ③ find $\frac{\partial J}{\partial w_{q1}^1} \Rightarrow$ from eq ③

$$(w_{q1}^1)^{\text{new}} \leftarrow (w_{q1}^1)^{\text{old}} - \eta \frac{\partial J}{\partial w_{q1}^1}$$

end

④ Repeat ② & ③ until some
stopping criterion
(loop/1000, iterations)

$\left\{ \begin{array}{l} A \\ A \\ A_0 \\ \boxed{A} \\ A \end{array} \right\} \rightarrow$

~~A~~

x

epoch $\div$ cycle over all the
examples $\Rightarrow$ data batch

$\gg \underline{100/1000}$

for epoch = 1 to 100

for data sim = 1 to 10,000

$$w \leftarrow w - \eta \frac{\partial F}{\partial w}$$